Doutorado em Matemática - IMPA

LECTURE NOTES ON DYNAMICS I

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1 Introduction

In this course we shall deal with *autonomous deterministic dynamical systems.* They can be used to represent or modelize any phenomenon that changes along the time in such a way that the current state of the system completely determines the future evolution and the law that drives this evolution does not change along the time. Mathematically, an **autonomous deterministic dynamical system** (a **dynamical system**, for short) is given by a set M, usually called the **phase space,** which represents the set of possible states of the system, a subsemigroup $\mathbb{T} \subset \mathbb{R}$ that represents the **time set,** and *"a law"* $\Phi\colon \mathbb{T} \times M \to M$ that determines the evolution of the system in such a way that given the current state of our system $x_0 \in M$, at time $t \in \mathbb{T}$ the corresponding state will be $x_t = \Phi(x_0, t)$. The autonomy of the system is reflected in the so called **flow** (or **semi-flow**) condition function Φ satisfies:

$$\Phi(t+s, x) = \Phi\big(t, \Phi(s, x)\big) = \Phi\big(t, \Phi(s, x)\big), \quad \forall t, s \in \mathbb{T},\ \forall x \in M. \tag{1}$$

We will consider two different kinds of dynamical systems: **discrete time** and **continuous time** ones, i.e. where $\mathbb{T} = \mathbb{N}_0$ and $\mathbb{T} = \mathbb{R}_{\geqslant 0}$, respectively. Moreover, we will also consider **invertible systems** where the current state does not just determine the future but the past as well. When dealing with invertible discrete dynamical systems, the time set will be $\mathbb{T} = \mathbb{Z}$, and in the case of invertible continuous time ones, we will have $\mathbb{T} = \mathbb{R}$.

The main goal of Dynamical Systems consists in describing the **orbits of the systems,** i.e. the sets

$$\mathcal{O}_\Phi(x) := \{\Phi(t, x) : t \in \mathbb{T}\}, \quad \forall x \in M.$$

Exercise 1.1 (Discrete time generator)**.** Given a discrete time dynamical system $\Phi\colon \mathbb{T} \times M \to M$, show that Φ satisfies the flow condition (1) if and only if there is $f\colon M \circlearrowleft$ such that

$$\Phi(n, x) = f^n(x), \quad \forall x \in M,\ \forall n \in \mathbb{N}_0,$$

where the iteration of f is inductively defined by $f^0 := id_M$ and $f^{n+1} = f \circ f^n$, for every $n \geqslant 0$. Show that Φ is invertible if and only if f is bijective and, in such a case, one can define $f^{-n} := \left(f^{-1}\right)^n$, for all $n \geqslant 0$.

On the other hand, one can find *infinitesimal generators* for continuous time smooth dynamical systems:

Exercise 1.2 (Continuous time generator)**.** Let M be a differentiable closed manifold (i.e. compact and boundaryless) and $\Phi\colon \mathbb{R} \times M \to M$ a continuous time dynamical system (that satisfies the flow condition 1). If we assume Φ is a C^2 map and we define $X(p) := \partial_t \Phi(t,p)\big|_{t=0} \in T_pM$, for every $p \in M$, then the function $\Phi(\cdot, x_0)\colon \mathbb{R} \to M$ is the unique solution of the *initial value problem*

$$\dot{x}(t) = X\big(x(t)\big), \quad x(0) = x_0,$$

for every $x_0 \in M$. In other words, the vector field X is the infinitesimal generator of the flow Φ.

As previously mentioned, the primary objective of Dynamical Systems is to understand the orbits of various systems, particularly their asymptotic behavior as time goes to infinity. From a historical perspective, there are three pivotal moments in the development of this mathematical theory. The first can be attributed to Isaac Newton, who laid the groundwork for Calculus and employed its principles to solve differential equations. During this era, the approach to studying a specific continuous-time dynamical system involved solving the corresponding differential equation, followed by an analysis of the obtained solutions.

The so-called *"restricted two-body problem"* is a remarkable question that was answered at that time:

Problem 1.3. The *"restricted two-body problem"* refers to a simplified model used to analyze the motion of two celestial bodies under the influence of their mutual gravitational attraction, where one body is significantly more massive than the other. In this framework, the more massive body is assumed to be stationary, while the second body is treated as a point mass that moves in the gravitational field created by the first one. This approach allows for the derivation of equations of motion for the second body without needing to account for its gravitational effects on the primary, simplifying the analysis. This model is particularly useful in celestial mechanics, where it can be applied to scenarios such as a satellite orbiting a planet or a comet passing by a star.

More precisely, the *restricted two-body problem* consider in describing the orbit of massive point p, whose location in the space at time $t \in \mathbb{R}$ is given by $x(t) \in \mathbb{R}^3$, moving in a central field force in $\mathbb{R}^3$ that is generated by a massive point of mass $M > 0$ which is fixed at the origin. According to Newton's laws, the trajectory $x(t) \in \mathbb{R}^3$ is the solution of the following initial value problem:

$$\begin{cases} \ddot{x}(t) = -\dfrac{GM}{\|x\|^3} x(t), \\ x(0) = x_0, \\ \dot{x}(0) = v_0, \end{cases} \tag{2}$$

where $G > 0$ is the universal constant of gravitation, $x_0 \in \mathbb{R}^3$ and $v_0 \in \mathbb{R}^3$ are the initial position and velocity, respectively.

Using classical *quadrature techniques* one can show that the solutions $x(t)$ of (2) are either straight lines through the origin (when x_0 and v_0 are collinear vectors in $\mathbb{R}^3$), or quadratic curves (i.e. ellipses, parabolas or hyperbolas).

The second pivotal moment in the development of Dynamical Systems can be attributed to Henri Poincaré. At the end of XIXth century it was already pretty clear that there were many ordinary differential equations that could not be solved with quadrature methods. A very simple and well-known example is the following one:

Example 1.1. *There is no* elementary (smooth) function $f\colon \mathbb{R} \to \mathbb{R}$ *such that*

$$f'(x) = e^{-x^2}, \quad \forall x \in \mathbb{R}.$$

Another example, which was fundamental in development of so called *qualitative theory of differential equations* is the 3-*body problem:*

Problem 1.4 (n-body problem)**.** The *n-body problem* refers to the challenge of predicting the individual motions of a group of n celestial objects that interact with each other through gravitational forces. The complexity arises from the fact that each body exerts a gravitational force on every other body, leading to a system of coupled differential equations. If $x_i(t) \in \mathbb{R}^3$ represents the position of the i-th object at time $t \in \mathbb{R}$ and $m_i > 0$ denotes its mass, then Newton's second law determines that these quantities are related by the following system of ordinary differential equations

$$\begin{cases} \ddot{x}_i(t) = \displaystyle\sum_{j \neq i} G \frac{m_j}{\|x_i(t) - x_j(t)\|^3} \big(x_i(t) - x_j(t)\big), \\ x_i(0) = x_{i,0}, \\ \dot{x}_i(0) = v_{i,0}, \end{cases} \tag{3}$$

for $1 \leqslant i \leqslant n$, and where $x_{i,0}, v_{i,0} \in \mathbb{R}^3$ denotes the position and the velocity of the i-th object at time $t = 0$, respectively.
It is pretty remarkable that, in general, the system of ODEs given by (3) cannot be solved by "quadrature methods" when $n \geqslant 3$. In fact, their solutions are not elementary functions in general.

The n-body problem, as outlined in Problem 1.4, served as a primary motivation for Henri Poincaré in developing the Geometric Theory of Differential Equations. Poincaré posited that if explicit solutions to a given differential equation cannot be readily obtained, one should at least strive to describe some interesting geometric properties of those solutions.

Indeed, many questions regarding the solutions of differential equations are more qualitative than quantitative, or more geometric than analytic. For instance, a fundamental qualitative inquiry concerning the system of ordinary differential equations in (3) is whether every solution is periodic. In some cases, even when solutions can be expressed using elementary functions, addressing certain qualitative questions based solely on the explicit formulas can prove to be quite challenging.

The third pivotal moment in the evolution of Dynamical Systems can be attributed to the schools of Steve Smale in the US and Aleksandr Andronov in the USSR. At this juncture, it became evident that the geometric properties of orbits, trajectories, or solutions to ODEs could change dramatically with variations in system parameters. For example, in Problem 1.4, there are $3n$ parameters—comprising the n masses, n initial positions, and n initial velocities. In certain instances, even minor changes to these parameters can lead to completely different geometric properties for the respective solutions. Thus, it becomes crucial to identify geometric properties of orbits that are robust or persistent under small perturbations of system parameters.

These three historical stages in the development of dynamical systems are cleverly (and provocatively) summarized by Étienne Ghys in his laudation for Artur Avila's Fields Medal: In Newton's time, you were given an ODE and tasked with finding its solutions. In Poincaré's era, you were given an ODE and asked to elucidate interesting geometric properties of its solutions. In Smale's time, you were given no ODE at all and challenged to say something interesting!

2 Some examples of Dynamical Systems results

A very first example of a result in Dynamical Systems might be the following classical

Theorem 2.1 (Banach fixed point theorem)**.** Let (M, d) be a complete metric space and $f\colon M \circlearrowleft$ be **uniform contraction,** i.e. f is a Lipschitz function and its **Lipschitz constant**

$$\lambda = \mathrm{Lip}(f) := \sup_{x \neq y} \frac{d\big(f(x), f(y)\big)}{d(x, y)} < 1.$$

Then there exists a unique point $p \in M$ such that $f(p) = p$ and it holds

$$\lim_{n \to +\infty} f^n(x) = p, \quad \forall x \in M.$$

Proof. Given any $x \in M$, one can show by induction on n that

$$d\big(f^n(x), f^{n+1}(x)\big) \leqslant \lambda^n d(x, f(x)), \quad \forall n \in \mathbb{N}_0.$$

So, for every $m, n \in \mathbb{N}$ it holds

$$\begin{aligned} d\big(f^m(x), f^{m+n}(x)\big) &\leqslant \sum_{i=0}^{n-1} d\big(f^{m+i}(x), f^{m+i+1}(x)\big) \\ &\leqslant \sum_{i=0}^{n-1} \lambda^{m+i} d\big(x, f(x)\big) \leqslant \frac{\lambda^m}{1-\lambda} d\big(x, f(x)\big). \end{aligned}$$

Hence, the sequence $\big(f^n(x)\big)_{n \geqslant 0}$ is Cauchy, and since M is complete, it converges to a point $p \in M$. By continuity of f, p is a fixed point of f. On the other hand, since f is a contraction, it has at most one fixed point. □

Exercise 2.1 (Perturbations of contractions)**.** Let $f\colon \mathbb{R}^d \circlearrowleft$ be a uniform contraction and $r > 0$ be arbitrary. We define the following balls

$$B_r^0(f) := \left\{ g \in C^0(\mathbb{R}^d, \mathbb{R}^d) : \|f - g\|_{C^0} := \sup_{x \in \mathbb{R}^d} \|f(x) - g(x)\| < r \right\},$$

and

$$B_r^L(f) := \left\{ g \in C^0(\mathbb{R}^d, \mathbb{R}^d) : \|f - g\|_L := \|f - g\|_{C^0} + \mathrm{Lip}(f - g) < r \right\}.$$

Show that given any $r > 0$, there is a $g \in B_r^0(f)$ which is not a contraction and such that $\mathrm{Fix}(g) := \{x \in \mathbb{R}^d : g(x) = x\}$ is an infinite set.
On the other hand, show that given the uniform contraction f, there exists $r_0 > 0$ such that every $g \in B_{r_0}^L(f)$ is a uniform contraction, and hence it exhibits a unique fixed point $p_g \in \mathbb{R}^d$. Finally, prove that given any $\varepsilon > 0$, there exists $\delta \in (0, r_0)$ such that

$$\|p_f - p_g\| < \varepsilon, \quad \forall g \in B_\delta^L(f).$$

Another important example is the following one:

Example 2.1 (Gradient flows)**.** *Let M be the unit sphere $\mathbb{S}^2 := \{v \in \mathbb{R}^3 : \|v\| = 1\}$ and consider the function $L\colon \mathbb{S}^2 \to \mathbb{R}$ given by*

$$L(x, y, z) := -z, \quad \forall (x, y, z) \in \mathbb{S}^2,$$

and the flow $\Phi\colon \mathbb{R} \times \mathbb{S}^2 \to \mathbb{S}^2$ generated by the vector field $X(v) := -\nabla L(v)$, for every $v \in \mathbb{S}^2$ and where ∇ denotes the gradient operator on $\mathbb{S}^2$.
One can easily show that $(0, 0, -1)$ and $(0, 0, 1)$ are the only singularities of X (i.e. the critical points of L), or equivalently, the fixed points of Φ, and it holds

$$\Phi^t(v) \to (0, 0, -1), \quad \text{and} \quad \Phi^{-t}(v) \to (0, 0, 1),$$

as $t \to +\infty$, for every $v \in \mathbb{S}^2 \backslash \{(0, 0, \pm 1)\}$.
Generalizing this example, one can consider an arbitrary compact differentiable manifold M endowed with a Riemannian structure $\langle \cdot, \cdot \rangle$ and a smooth function $L\colon M \to \mathbb{R}$. Then we can define the gradient of L as the only vector field ∇L such that

$$YL = dL(Y) = \langle \nabla L, Y \rangle,$$

for every vector field Y on M.
So, if we consider the flow $\Phi\colon \mathbb{R} \times M \to M$ induced by the vector field $X := -\nabla L$, one can easily check that

$$\partial_t (L \circ \Phi^t)(p)\big|_{t=0} = -\|\nabla L(p)\|^2 = -\langle \nabla L(p), \nabla L(p) \rangle, \quad \forall p \in M.$$

That means, given any point $p \in M$ the map $\mathbb{R} \ni t \mapsto L \circ \Phi^t(p) \in \mathbb{R}$ is not increasing, and it is strictly decreasing if and only if p is a regular point for L is a regular point of L, i.e. $dL_p\colon T_pM \to \mathbb{R}$ is not identically zero.
Moreover, if L has a finite number of critical points (i.e. points where the derivative dL vanishes), then one can show that for every $p \in M$, there exist critical points of L $\alpha_p, \omega_p \in M$ such that

$$\Phi^{-t}(p) \to \alpha_p, \qquad \Phi^t(p) \to \omega_p,$$

as $t \to +\infty$.

3 The Fundamental Theorem of Dynamical Systems

Motivated by Example 2.1, we would like to find functions decreasing along the orbits of arbitrary topological dynamical systems.

Definition 3.1 (Lyapunov function)**.** Given a continuous dynamical systems $f\colon M \circlearrowleft$ on a metric space M, a *Lyapunov function* for f is a continuous function $L\colon M \to \mathbb{R}$ such that

$$L\big(f(x)\big) \leqslant L(x), \quad \forall x \in M.$$

Of course, according to this definition every constant function is a Lyapunov one for every topological dynamical system, which does not seem to be very interesting. So, we are going to look for the "best possible" Lyapunov function.

3.1 Different forms of recurrence

It would be desirable to find a Lyapunov function such that $L\big(f(x)\big) < L(x)$ for every $x \in M$. However this is clearly impossible when f has, for instance, a **fixed point,** i.e. a point $p \in M$ such that $f(p) = p$. The set of fixed points of f will be denoted by $\mathrm{Fix}(f)$.

More generally, we write $p \in \mathrm{Per}(f)$ and say that p is a **periodic point** when there is $n \geqslant 1$ such that $f^n(p) = p$. We define $\mathrm{Per}(f) := \bigcup_{n\geqslant 1} \mathrm{Fix}(f^n)$.

Another form of recurrence that represents an obstruction for the existence of a Lyapunov functions satisfying the strict inequality for every $p \in M$ is the following: we say that a point $p \in M$ is **positively recurrent** for f when there exists strictly increasing sequence of natural numbers $n_j \uparrow +\infty$ such that $f^{n_j}(p) \to p$, as $n_j \to +\infty$. The set of positively recurrent points of f shall be denoted by $\mathrm{Rec}^+(f)$. When $f\colon M \circlearrowleft$ is bijective, one says that $p \in M$ is **negatively recurrent** for f when it is positively recurrent for f^{-1}. The set of negatively recurrent points will be denoted by $\mathrm{Rec}^-(f)$.

Given a point $p \in M$, we define its **ω-limit set** by

$$\omega_f(p) := \{x \in M : \exists n_j \uparrow \infty,\ f^{n_j}(p) \to x,\ \text{as } n_j \to +\infty\},$$

and the **positive limit set** of f by

$$L^+(f) := \bigcup_{p\in M} \omega_f(p).$$

When f is invertible, we also define the **α-limit set** of $p \in M$ by $\alpha_f(p) := \omega_{f^{-1}}(p)$, and the so called **limit set** of f by

$$L(f) := \bigcup_{p\in M} \alpha_f(p) \cup \omega_f(p).$$

We say that a point $p \in M$ is **non-wandering** when for every neighborhood U of p, there exists $n \geqslant 1$ such that $U \cap f^{-n}(U) \neq \varnothing$. The set of non-wandering points for f is usually denoted by $\Omega(f)$.

We say that a subset $K \subset M$ is **f-invariant** when $f(K) = K$, and it is said to be **totally f-invariant** when $f(K) = K = f^{-1}(K)$. Observe that if $f\colon M \circlearrowleft$ is bijective, then $K \subset M$ is totally f-invariant if and only if it is f-invariant.

Exercise 3.1. Let M denote a compact metric space and $f\colon M \circlearrowleft$ be a homeomorphism. Show that

$$\mathrm{Fix}(f) \subset \mathrm{Per}(f) \subset \mathrm{Rec}^+(f) \subset L(f) \subset \Omega(f),$$

and all these sets are f-invariant.

1. Show that $\mathrm{Fix}(f)$ and $\Omega(f)$ are compact.
2. Show that $\mathrm{Rec}^+(f) \cap \mathrm{Rec}^-(f) \neq \varnothing$. (Hint: show that there exists a non-empty compact subset $K \subset M$ which is **minimal** for f, i.e. K is f-invariant and it does not contain any proper compact f-invariant subset).
3. Show that if $L\colon M \to \mathbb{R}$ is a Lyapunov function for f and $p \in \Omega(f)$, then it holds $L\big(f(p)\big) = L(p)$.

Finally we introduce the notion of *chain recurrence.* Given a homeomorphism $f\colon M \circlearrowleft$ on the compact metric space (M, d) and an arbitrary $\varepsilon > 0$, a sequence $(x_n)_{n\in\mathbb{Z}}$ of points of M is said to be an **ε-chain** or **ε-pseudo-orbit** when

$$d\big(f(x_i), x_{i+1}\big) < \varepsilon, \quad \forall n \in \mathbb{Z}.$$

A point $p \in M$ is said to be **chain recurrent** when for every $\varepsilon > 0$ there exists a periodic ε-chain $(x_n)_{n\in\mathbb{Z}}$ with $p = x_0$. The set of chain recurrent points is denoted by $\mathscr{R}(f)$.

Exercise 3.2. Show that $\mathscr{R}(f)$ is a compact f-invariant set and $\Omega(f) = \Omega(f^{-1}) \subset \mathscr{R}(f) = \mathscr{R}(f^{-1})$.

Given any $p \in M$ and any $\varepsilon > 0$, we define the **ε-chainable set from** p as the set

$$\mathscr{R}_f(p, \varepsilon) := \{x \in M : \exists (x_n)_{n\in\mathbb{Z}}\ \varepsilon\text{-chain}, \exists k \in \mathbb{N}, x_0 = p,\ x_k = x\},$$

and the **chainable set from** p by

$$\mathscr{R}_f(p) := \bigcap_{\varepsilon>0} \mathscr{R}_f(p, \varepsilon).$$

So, we have $p \in \mathscr{R}(f)$ if and only if $p \in \mathscr{R}_f(p)$.

There is a fundamental and elementary notion that will play a crucial role in the understanding of the orbit structure of a given dynamical system, which is the concept of *trapping region.* We say that a subset $U \subset M$ is a **trapping region** for f when it is open and $f\big(\overline{U}\big) \subset U$.

Observe that if $U \subset M$ is a trapping region for f, then the set $M \backslash \overline{U}$ is a trapping region for f^{-1}; and reciprocally, if $M \backslash \overline{U}$ is a trapping for f^{-1}, then $\mathrm{int}(\overline{U})$ is a trapping region for f.

Given a trapping region U for f, we can define the **dual pair of attractor/repellor** asso-

ciated to U as the pair of compact f-invariant sets

$$A_U^+ := \bigcap_{n\geqslant 0} f^n\big(\overline{U}\big), \quad \text{and} \quad A_U^- := \bigcap_{n\geqslant 0} f^{-n}\big(M\backslash U\big).$$

Exercise 3.3. Show that A_U^+ and A_U^- are compact, f-invariant and contained in U and $M\backslash\overline{U}$, respectively. Moreover, they are non-empty if and only if U and $M\backslash\overline{U}$ are non-empty.

We have the following fundamental result relating chain recurrence and trapping regions:

Proposition 3.1. Let $f\colon M \circlearrowleft$ be a homeomorphism on the compact metric space (M,d). Then, given any point $p\in M$ and any $\varepsilon>0$, the ε-chainable set from p is a trapping region for f.

Proof. Let us start showing that $\mathscr{R}_f(p,\varepsilon)$ is open. To do that, let us inductively defined

$$U_1 := B_\varepsilon(p) = \{y\in M : d(x,y)<\varepsilon\},$$

and

$$U_n := \bigcup_{x\in U_{n-1}} B_\varepsilon\big(f(x)\big), \quad \forall n\geqslant 2.$$

One can easily check that U_n is open for every $n\geqslant 1$ and $\mathscr{R}_f(p,\varepsilon) = \bigcup_{n\geqslant 1} U_n$. So, $\mathscr{R}_f(p,\varepsilon)$ is open.

By the very same definition, it is clear that $f\big(\mathscr{R}_f(p,\varepsilon)\big)\subset \mathscr{R}_f(p,\varepsilon)$. Now, let y be an arbitrary point of $\overline{\mathscr{R}_f(p,\varepsilon)}$. Since M is compact and f is continuous, there is $\delta>0$ such that $d\big(f(x_1),f(x_2)\big)<\varepsilon$, whenever $d(x_1,x_2)<\delta$. Then, there is $x\in\mathscr{R}_f(p,\varepsilon)\cap B_\delta(y)$. So there is a finite segment of ε-pseudo orbit $x_0=p,x_1,\dots,x_n=x$. Since $x\in B_\delta(y)$, we conclude $x_0=p,x_1,\dots,x_n=x,x_{n+1}:=f(y)$ is a finite segment of ε-pseudo orbit, too. Hence, $f(y)\in\mathscr{R}_f(p,\varepsilon)$, and we conclude that $f\big(\overline{\mathscr{R}_f(p,\varepsilon)}\big)\subset\mathscr{R}_f(p,\varepsilon)$. $\square$

Exercise 3.4. Let $f\colon M\circlearrowleft$ be a homeomorphism on the compact metric space (M,d) and $U\subset M$ be a trapping region for f. Show that

$$\mathscr{R}(f)\subset A_U^+\sqcup A_U^-,$$

where $\sqcup$ denotes the disjoint union and A_U^+, A_U^- is the dual pair of attractor/repellor associated to U.

Now we can define the notion of **chain recurrence classes.** If $f\colon M\circlearrowleft$ is a homeomorphism on the compact metric space (M,d) and $x,y\in\mathscr{R}(f)$, we say that they are **chain equivalent** when $y\in\mathscr{R}_f(x)$ and $x\in\mathscr{R}_f(y)$, i.e. $x,y\in\mathscr{R}_f(x,\varepsilon)\cap\mathscr{R}_f(y,\varepsilon)$, for every $\varepsilon>0$.

Then we have the following

Exercise 3.5 (Chain-recurrence classes)**.** If f and M are as above, let us show that *chain equivalence* determines an equivalence relation on $\mathscr{R}(f)$. Their equivalence classes are called **chain-recurrence classes.** Show that every chain-recurrent class is compact and f-invariant.
Moreover, two points $x, y \in \mathscr{R}(f)$ are chain equivalent if and only if either $x, y \in A_U^+$, or $x, y \in A_U^-$, for every trapping region U for f.

Exercise 3.6. If f and M are as above, show that the following set is at mots countable:

$$\left\{(A_U^+, A_U^-) : U \subset M \text{ is a trapping region for } f\right\}.$$

Using all previous exercises and results, one can easily prove the so-called "Fundamental Theorem of Dynamical Systems" due to Conley [Con78]:

Theorem 3.2 (Existence of complete Lyapunov functions)**.** Given a homeomorphism $f\colon M \circlearrowleft$ on the compact metric space (M, d), there exists a *complete Lyapunov function* for f, i.e. a continuous function $L\colon M \to \mathbb{R}$ such that the following conditions hold:

1. $L\big(f(x)\big) \leqslant L(x)$, for every $x \in M$,
2. $L\big(f(x)\big) = L(x)$ if and only if $x \in \mathscr{R}(f)$,
3. $L(x) = L(y)$, with $x, y \in \mathscr{R}(f)$ if and only if x and y are chain equivalent, and,
4. $L\big(\mathscr{R}(f)\big)$ is a compact totally disconnected set, i.e. it does not contain any open interval.

4 Dynamics on the real line

4.1 Affine maps of $\mathbb{R}$

For each $a, b \in \mathbb{R}$, we can define the affine map $f_{a,b}\colon \mathbb{R} \circlearrowleft$ by $f_{a,b}(x) := ax + b$, for every $x \in \mathbb{R}$. We are going to describe the orbit structure of $f_{a,b}$ for every choice of $a, b \in \mathbb{R}$.

If $a = 1$ and $b = 0$, then $f_{1,0} = id_{\mathbb{R}}$. If $a = 1$ and $b > 0$, then $f_{1,b}^n(x) \to +\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}$. If $a = 1$ and $b < 0$, then $f_{1,b}^n(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}$.

If $a \neq 1$ and $b \in \mathbb{R}$ is arbitrary, then $p_{a,b} = -\frac{b}{a-1}$ is the unique fixed point of $f_{a,b}$. Moreover, if $|a| < 1$, then $f_{a,b}^n(x) \to p_{a,b}$, as $n \to +\infty$, for every $x \in \mathbb{R}$. If $|a| > 1$, then $\left|f_{a,b}^n(x)\right| \to +\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}\backslash\{p_{a,b}\}$.

Finally, if $a = -1$ and $b \in \mathbb{R}$ is arbitrary, then we have $f_{-1,b}^2 = id_{\mathbb{R}}$, and $\mathrm{Fix}(f_{-1,b}) = \{p_{-1,b}\}$ and $\mathrm{Per}(f_{-1,b}) = \mathbb{R}$.

4.2 The quadratic family

The dynamics of quadratic polynomials in $\mathbb{R}$ is much more interesting than the one of affine maps. Now, for each triple of real numbers $a, b, c \in \mathbb{R}$, with $a \neq 0$, we can define the

quadratic map $f_{a,b,c}\colon \mathbb{R} \circlearrowleft$ by $f_{a,b,c}(x) := ax^2 + bx + c$, for every $x \in \mathbb{R}$.

First observe that $\mathrm{Fix}(f_{a,b,c}) = \varnothing$ if and only if $(b-1)^2 - 4ac < 0$. In such a case, when $a > 0$, it holds $f^n_{a,b,c}(x) \to +\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}$. When $a < 0$, it holds $f^n_{a,b,c}(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}$.

So, from now on we are going to assume that $(b-1)^2 - 4ac \geqslant 0$ (with $a \neq 0$), and hence $\mathrm{Fix}(f_{a,b,c}) \neq \varnothing$.

We are going to introduce a fundamental concept in Dynamical Systems, which corresponds to the notion of isomorphism in the category of topological dynamical systems:

Definition 4.1 (Conjugacy)**.** Given two metric spaces M, N and two continuous maps $f\colon M \circlearrowleft$ and $g\colon N \circlearrowleft$, we say that f and g are **topologically conjugate** when there exists a homeomorphism $h\colon M \to N$ such that

$$h \circ f = g \circ h.$$

When M and N are differentiable manifolds and $f\colon M \circlearrowleft$ and $g\colon N \circlearrowleft$ are C^r maps, with $r \in \mathbb{N} \cup \{\infty\}$, we say that f and g are **C^s-conjugate** when there exists a C^s-diffeomorphism $h\colon M \to N$, with $s \in \mathbb{N} \cup \{\infty\}$ and $s \leqslant r$.

To simplify the study of the dynamics of quadratic maps, we are going to show that every quadratic map with fixed points is affinely conjugate to a quadratic map of the form $Q_\mu(x) := \mu x(1-x)$, for some $\mu > 0$.

In order to do that, first suppose that $a > 0$. If we consider the affine map $h\colon \mathbb{R} \ni x \mapsto -x$, we easily see that $h \circ f_{a,b,c} = f_{-a,b,-c} \circ h$. So, from now on we can assume that $a < 0$.

Then, the vertex of the parabola defined by $f_{a,b,c}$ is at the point $v = -\frac{b}{2a}$ and since we are assuming that $a < 0$ and $(b-1)^2 - 4ac \geqslant 0$, we know that $f_{a,b,c}$ has a fixed point $p \in \mathbb{R}$ with $p < v$. So, there is an affine map $h\colon \mathbb{R} \ni x \mapsto \alpha x + \beta$ such that $h(0) = p$ and $h(1/2) = v$.

Problem 4.1. If $f_{a,b,c}$, v, p and h are as above, show that there is $\mu > 0$ such that $h \circ f_{a,b,c} = Q_\mu \circ h$.

So, from now on we shall concentrate on the dynamics of the family of quadratic maps $\{Q_\mu : \mu > 0\}$, which is called the **quadratic family.** For each $\mu > 0$, observe that

$$p_\mu := \frac{\mu - 1}{\mu},$$

is a fixed point of Q_μ, and hence $\mathrm{Fix}(Q_\mu) = \{0, p_\mu\}$.

We have the following result describing the dynamics in the neighborhood of the fixed points:

Proposition 4.1. Let $I \subset \mathbb{R}$ be an open interval and $f\colon I \circlearrowleft$ be a C^1 map. If $p \in I$ is a fixed point of f and $|f'(p)| \neq 1$, then the following dichotomy holds:

- If $|f'(p)| < 1$, then there is $\delta > 0$ such that $f^n(x) \to p$, as $n \to +\infty$, for every $x \in (p-\delta, p+\delta) \subset I$;
- if $|f'(p)| > 1$, then there is $\delta > 0$ such that for every $x \in I$ with $0 < |x-p| < \delta$, there is $n \in \mathbb{N}$ such that $|f^n(x) - p| \geqslant \delta$.

Proof. In the first case, since $|f'(p)| < 1$ and I is open, there is $\delta > 0$ and $\lambda \in (0,1)$ such that $|f'(x)| < \lambda$, for every $x \in (p-\delta, p+\delta) \subset I$. By the mean value theorem, given any $x \in (p-\delta, p+\delta)$, it holds

$$|f(x) - p| \leqslant \lambda\,|x-p| < \lambda\delta < \delta.$$

So, f restricted to $(p-\delta, p+\delta)$ is a uniform contraction, and by the Banach fixed point theorem, $f^n(x) \to p$, as $n \to +\infty$, for every $x \in (p-\delta, p+\delta)$.

Now, let us assume that $|f'(p)| > 1$. There is $\delta > 0$ and $\lambda > 1$ such that $|f'(x)| > \lambda$, for every $x \in (p-\delta, p+\delta)$. By the mean value theorem, given any $x \in (p-\delta, p+\delta)$, if $f^i(x) \in (p-\delta, p+\delta)$, for every $i \in \{0, 1, \ldots, m\}$, then it holds

$$\delta > |f^m(x) - p| \geqslant \lambda^n\,|x-p|\,.$$

So, $m \leqslant \log(\delta/\,|x-p|)/\log\lambda$ and the proof is completed. □

Definition 4.2. A fixed point p of a C^1 map $f\colon I \circlearrowleft$ is said to be **hyperbolic** when $|f'(p)| \neq 1$. If $|f'(p)| < 1$, then p is called an **attractor**, and if $|f'(p)| > 1$, then p is called a **repeller**.

Returning to the quadratic family, we have that $Q'_\mu(x) = -2\mu x + \mu$, and thus, $Q'_\mu(0) = \mu$ and $Q'_\mu(p_\mu) = 2 - \mu$.

So, if $\mu \in (0,1)$, $p_\mu < 0$, p_μ is a repeller and 0 is an attractor. In such a case, one can easily check that $Q^n_\mu(x) \to 0$, as $n \to +\infty$, for every $x \in (p_\mu, q_\mu)$, where q_μ is the unique point of $(p_\mu, +\infty)$ such that $Q_\mu(q_\mu) = p_\mu$; and $Q^n_\mu(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}\backslash[p_\mu, q_\mu]$.

If $\mu = 1$, then $p_\mu = 0$ and 0 turns to be a non-hyperbolic fixed point. In such a case, one can easily check that $Q^n_1(x) \to 0$, as $n \to +\infty$, for every $x \in (0,1)$, and $Q^n_1(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}\backslash[0,1]$.

Exercise 4.2. Show that if $\mu \in (1,3)$, then $p_\mu \in (0,1)$ is a hyperbolic attractor, while 0 is a repeller, $Q^n_\mu(x) \to p_\mu$, as $n \to +\infty$, for every $x \in (0,1)$; and $Q^n_\mu(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}\backslash[0,1]$.

If $\mu = 3$, the fixed point p_μ turns to be a non-hyperbolic attractor, and one can easily check that $Q^n_3(x) \to p_3$, as $n \to +\infty$, for every $x \in (0,1)$; and $Q^n_3(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}\backslash[0,1]$.

If $\mu \in (3,4)$, then both fixed points of Q_μ are repeller, the interval $[0,1]$ is a totally invariant set for Q_μ and the dynamics of Q_μ on $[0,1]$ is much more complicated (see for instance [Dev89] for more details on this topic).

And now, we shall concentrate on the case $\mu > 4$, paying special attention to the case $\mu > \sqrt{5}+2$. When $\mu > 4$, the interval $[0,1]$ is not invariant for Q_μ anymore. In fact, $Q_\mu(1/2) = \mu/4 > 1$ and hence there are two points r_μ, s_μ, with $0 < r_\mu < 1/2 < s_\mu < 1$ such that $Q_\mu(r_\mu) = Q_\mu(s_\mu) = 1$. Since $Q_\mu^n(x) \to -\infty$, as $n \to +\infty$, for every $x \in \mathbb{R}\backslash[0,1]$, we conclude that $Q_\mu^n(x) \to -\infty$, as $n \to +\infty$, for every $x \in (r_\mu, s_\mu)$. This implies that

$$[0,1] \supsetneq Q_\mu^{-1}([0,1]) \supsetneq Q_\mu^{-2}([0,1]) \supsetneq \cdots \supsetneq Q_\mu^{-n}([0,1]) \supsetneq \cdots,$$

and hence

$$\Lambda_\mu := \bigcap_{n\in\mathbb{N}} Q_\mu^{-1}\big([0,1]\big), \tag{4}$$

is a compact, non-empty totally invariant set for Q_μ, which does not coincide with the interval $[0,1]$.

In fact, $\Lambda_\mu \subset I_\mu^1 \sqcup I_\mu^2$, where $I_\mu^1 := [0, r_\mu]$ and $I_\mu^2 := [s_\mu, 1]$. So, if we write $[\![k]\!] := \{1, 2, \ldots, k\}$, for any $k \in \mathbb{N}$, we can define the so-called **itinerary map** by $h\colon \Lambda_\mu \to [\![2]\!]^{\mathbb{N}_0}$ where

$$h(x)_n := \begin{cases} 1, & \text{if } Q_\mu^n(x) \in I_\mu^1, \\ 2, & \text{if } Q_\mu^n(x) \in I_\mu^2, \end{cases} \quad \forall n \in \mathbb{N}_0.$$

In general, given any $k \in \mathbb{N}$ with $k \geqslant 2$, if we endowed the space $[\![k]\!]^{\mathbb{N}_0}$ with the product topology, then by Thychonov theorem we get that $[\![k]\!]^{\mathbb{N}_0}$ is compact. Moreover, its topology is metrizable and compatible with the topology induced by the metric $d\colon [\![k]\!]^{\mathbb{N}_0} \to [0,1]$ given by

$$d\big((i_n),(j_n)\big) := 2^{-\min\{n\in\mathbb{N}_0 : i_n \neq j_n\}}, \quad \forall (i_n),(j_n) \in [\![k]\!]^{\mathbb{N}_0},$$

where we are assuming that $\min \varnothing = +\infty$ and hence $d\big((i_n),(j_n)\big) = 0$, if $i_n = j_n$, for every $n \in \mathbb{N}_0$.

We need the following

Definition 4.3 (Cantor spaces)**.** A metrizable topological space K is said to be a **Cantor space** when it is compact, **perfect** (i.e. there is no isolated point) and **totally disconnected** (i.e. if $C \subset K$ is connected and non-empty, then it is a singleton).

Exercise 4.3. Show that $[\![k]\!]^{\mathbb{N}_0}$ is a Cantor space, for every $k \in \mathbb{N}$ with $k \geqslant 2$.

With this topology, it is easy to check that the itinerary map $h\colon \Lambda_\mu \to [\![2]\!]^{\mathbb{N}_0}$ that we defined above is, indeed, continuous. Moreover, since $\mu > \sqrt{5}+2$, there is $\lambda > 1$ such that $\big|Q_\mu'(x)\big| > \lambda$, for every $x \in I_\mu^1 \cup I_\mu^2$. So, given any two distinct points $x, y \in \Lambda_\mu$, there is $n \in \mathbb{N}_0$ such that $Q_\mu^n(x)$ and $Q_\mu^n(y)$ belong to different intervals of the form I_μ^1 or I_μ^2. Hence, we conclude that the itinerary map $h\colon \Lambda_\mu \to [\![2]\!]^{\mathbb{N}_0}$ is injective.

Moreover, the map $Q_\mu\big|_{I_\mu^i}\colon I_\mu^i \to [0,1]$ is a diffeomorphisms for $i = 1,2$. So, they admit well defined inverse maps that we will denote by $g_i\colon [0,1] \to I_\mu^i$, for $i = 1,2$, which are

contraction with Lipschitz constants smaller or equal than λ^{-1}. Then, given any sequence $(i_n) \in [\![2]\!]^{\mathbb{N}_0}$, we can define the point $x \in \Lambda_\mu$ by the following intersection:

$$\{x\} := \bigcap_{n \in \mathbb{N}_0} g_{i_0} \circ g_{i_1} \circ \cdots \circ g_{i_n}([0,1]).$$

One can easily check that $h(x) = (i_n)$, and hence the itinerary map $h\colon \Lambda_\mu \to [\![2]\!]^{\mathbb{N}_0}$ is continuous, injective and surjective, and thus it is a homeomorphism.

On the other hand, for any $k \in \mathbb{N}_0$ larger than 1, we can define the **shift map** $\sigma\colon [\![k]\!]^{\mathbb{N}_0} \circlearrowleft$ by

$$\sigma\big((i_n)\big) := (j_n), \text{ with } j_n := i_{n+1}, \quad \forall (i_n) \in [\![k]\!]^{\mathbb{N}_0}, \ \forall n \in \mathbb{N}_0.$$

The main elementary properties of the shift map are given by the following

Exercise 4.4. If $k \in \mathbb{N}$ with $k \geqslant 2$, and $\sigma\colon [\![k]\!]^{\mathbb{N}_0} \circlearrowleft$ denotes the shift map, then shw that the following properties hold:

1. σ is continuous and surjective;
2. the set $\mathrm{Per}(\sigma)$ is dense in $[\![k]\!]^{\mathbb{N}_0}$;
3. σ is **topologically transitive,** i.e. there is a point $x \in [\![k]\!]^{\mathbb{N}_0}$ such that $\{\sigma^n(x) : n \in \mathbb{N}_0\}$ is dense in $[\![k]\!]^{\mathbb{N}_0}$;
4. for every $x \in [\![k]\!]^{\mathbb{N}_0}$, the set
$$\big\{y \in [\![k]\!]^{\mathbb{N}_0} : \exists n \in \mathbb{N}, \ \sigma^n(y) = x\big\}$$
is dense in $[\![k]\!]^{\mathbb{N}_0}$;
5. for every $U \subset [\![k]\!]^{\mathbb{N}_0}$ open and non-empty, there is $n \in \mathbb{N}$ such that $\sigma^n(U) = [\![k]\!]^{\mathbb{N}_0}$:

5 Dynamics of circle maps

Our preferred model for the circle is the quotient space $\mathbb{T} := \mathbb{R}/\mathbb{Z}$. The natural quotient map $\pi\colon \mathbb{R} \to \mathbb{T}$ given by $\pi(x) := [x] = x + \mathbb{Z}$, for every $x \in \mathbb{R}$, can be used to endow $\mathbb{T}$ with the quotient topology, which is, by the definition, the finest topology on $\mathbb{T}$ such that π is continuous. This topology is metrizable and compatible with the metric $d\colon \mathbb{T} \times \mathbb{T} \to [0, 1/2]$ given by

$$d([x],[y]) := \min\{|\tilde{x} - \tilde{y}| : \tilde{x} \in [x], \ \tilde{y} \in [y]\} \quad \forall x, y \in \mathbb{R}.$$

One can easily check that $\pi\colon \mathbb{R} \to \mathbb{T}$ is a covering map, i.e. for every $[x] \in \mathbb{T}$ there is an open neighborhood U of $[x]$ such that $\pi^{-1}(U)$ is a disjoint union of open intervals of $\mathbb{R}$, each of them homeomorphic to U via π, and since $\mathbb{R}$ is simply connected, this covering map is the universal covering of $\mathbb{T}$. So, we have the following

Proposition 5.1. Given any continuous map $f\colon \mathbb{T} \circlearrowleft$, any $x_0 \in \mathbb{T}$, any $\tilde{x}_0 \in \mathbb{R}$ with $\pi(\tilde{x}_0) = x_0$, and any $\tilde{y}_0 \in \mathbb{R}$ such that $\pi(\tilde{y}_0) = f(x_0)$, there exists a unique continuous map $\tilde{f}\colon \mathbb{R} \circlearrowleft$ such that $\pi \circ \tilde{f} = f \circ \pi$ and $\tilde{f}(\tilde{x}_0) = \tilde{y}_0$. Moreover, if f is a homeomorphism, then $\tilde{f}$ is a homeomorphism, too.

As an easy consequence of the lifting property, we have that if $\tilde{f}_1, \tilde{f}_2\colon \mathbb{R} \circlearrowleft$ are two lifts of the same continuous map $f\colon \mathbb{T} \circlearrowleft$, then there is $n \in \mathbb{Z}$ such that $\tilde{f}_1(x) = \tilde{f}_2(x) + n$, for every $x \in \mathbb{R}$.

On the other hand, we have the following

Proposition 5.2. If $f\colon \mathbb{T} \circlearrowleft$ is a continuous, then there is an integer $d \in \mathbb{Z}$, called the **degree of** f, such that given any lift $\tilde{f}\colon \mathbb{R} \circlearrowleft$ of f, it holds $\tilde{f}(x+1) = \tilde{f}(x) + d$, for every $x \in \mathbb{R}$. Moreover, the map

$$\Delta_{\tilde{f}}(\tilde{x}) := \tilde{f}(\tilde{x}) - d\tilde{x}, \quad \forall \tilde{x} \in \mathbb{R},$$

is continuous and $\mathbb{Z}$-periodic, i.e. $\Delta_{\tilde{f}}(\tilde{x} + n) = \Delta_{\tilde{f}}(\tilde{x})$, for every $\tilde{x} \in \mathbb{R}$ and every $n \in \mathbb{Z}$.

Proof. Given any $\tilde{x} \in \mathbb{R}$, since $\pi\big(\tilde{f}(\tilde{x}+1)\big) = f\big(\pi(\tilde{x}+1)\big) = f\big(\pi(\tilde{x})\big) = \pi\big(\tilde{f}(\tilde{x})\big)$, we conclude that there is $d \in \mathbb{Z}$ such that $\tilde{f}(\tilde{x}+1) = \tilde{f}(\tilde{x}) + d$. Since $\tilde{f}$ is a continuous map, we conclude that the integer d does not depend on the choice of $\tilde{x} \in \mathbb{R}$. Moreover, by our remark above about different lifts of the same map, we conclude that d does not depend on the choice of the lift $\tilde{f}$ either, and the first part of the proposition is proved.

On the other hand, to prove that the map $\Delta_{\tilde{f}}$ is $\mathbb{Z}$-periodic, it is enough to check that

$$\Delta_{\tilde{f}}(\tilde{x}+1) = \tilde{f}(\tilde{x}+1) - d(\tilde{x}+1) = \tilde{f}(\tilde{x}) + d - d\tilde{x} - d = \Delta_{\tilde{f}}(\tilde{x}),$$

for every $\tilde{x} \in \mathbb{R}$. $\square$

Exercise 5.1. Given two continuous maps $f, g\colon \mathbb{T} \circlearrowleft$, we write d_f and d_g for the degrees of f and g, respectively, and $d_{f\circ g}$ for the degree of $f \circ g$. Show that

$$d_{f\circ g} = d_f \cdot d_g.$$

Since the degree of the identity map $id_{\mathbb{T}}$ is clearly 1, by the previous exercise we conclude that if $f\colon \mathbb{T} \circlearrowleft$ is a homeomorphism, then its degree is either 1 or -1.

5.1 Dynamics of circle homeomorphisms

By the last remark of the previous section, we know that the group of homeomorphisms of the circle $\mathbb{T}$, that shall be denoted by $\mathrm{Homeo}_(\mathbb{T})$, can be decomposed as the disjoint union of two subsets

$$\mathrm{Homeo}(\mathbb{T}) = \mathrm{Homeo}_+(\mathbb{T}) \sqcup \mathrm{Homeo}_-(\mathbb{T}),$$

where we are writing $\mathrm{Homeo}_+(\mathbb{T}) := \{f \in \mathrm{Homeo}(\mathbb{T}) : \text{the degree of } f \text{ is } 1\}$ and $\mathrm{Homeo}_-(\mathbb{T}) := \{f \in \mathrm{Homeo}(\mathbb{T}) : \text{the degree of } f \text{ is } -1\}$.

The elements of $\mathrm{Homeo}_+(\mathbb{T})$ shall be called **orientation-preserving** homeomorphisms, while the elements of $\mathrm{Homeo}_-(\mathbb{T})$ shall be called **orientation-reversing** homeomorphisms. One can easily check that any lift of an element of $\mathrm{Homeo}_+(\mathbb{T})$ is a strictly increasing homeomorphism of $\mathbb{R}$, while any lift of an element of $\mathrm{Homeo}_-(\mathbb{T})$ is a strictly decreasing homeomorphism of $\mathbb{R}$.

Exercise 5.2 (Dynamics of orientation-reversing circle homeomorphisms)**.** Given any $f \in \mathrm{Homeo}_-(\mathbb{T})$, show that $\mathrm{Fix}(f)$ contains exactly two points and $\mathrm{Per}(f) = \mathrm{Fix}(f^2)$, where $f^2 \in \mathrm{Homeo}_+(\mathbb{T})$ is an orientation-preserving circle homeomorphism.

So, let us start the study of orientation-preserving circle homeomorphisms. We have the following

Exercise 5.3. A homeomorphism $\tilde{f} \in \mathrm{Homeo}(\mathbb{R})$ is a lift of an element of $\mathrm{Homeo}_+(\mathbb{T})$ if and only if $\tilde{f}$ is continuous, strictly increasing and $\tilde{f}(x+1) = \tilde{f}(x) + 1$, for every $x \in \mathbb{R}$. In such a case, $\tilde{f}$ commutes with every integer translation, i.e. $\tilde{f} \circ T_m = T_m \circ \tilde{f}$, for every $m \in \mathbb{Z}$, where $T_m : \mathbb{R} \ni x \mapsto x + m \in \mathbb{R}$, and the function

$$\Delta_{\tilde{f}}(x) := \tilde{f}(x) - x, \quad \forall x \in \mathbb{R},$$

is continuous and $\mathbb{Z}$-periodic, i.e. $\Delta_{\tilde{f}}(x+n) = \Delta_{\tilde{f}}(x)$, for every $x \in \mathbb{R}$ and every $n \in \mathbb{Z}$. In such a case, we can consider the function $\Delta_{\tilde{f}}$ can be considered as an element of $C^0(\mathbb{T}, \mathbb{R})$, and is usually called the **displacement function** of $\tilde{f}$.

From now on, we shall write

$$\widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T}) := \left\{ \tilde{f} \in \mathrm{Homeo}(\mathbb{R}) : \tilde{f} \text{ is a lift of an } f \in \mathrm{Homeo}_+(\mathbb{T}) \right\}.$$

Exercise 5.4. Show that $\widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ is a subgroup of $\mathrm{Homeo}(\mathbb{R})$. On the other hand, show that given $\tilde{f}, \tilde{g} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$, it holds

$$\Delta_{\tilde{f} \circ \tilde{g}} = \Delta_{\tilde{f}} \circ g + \Delta_{\tilde{g}}.$$

Use this fact to show that

$$\Delta_{\tilde{f}^n} = \sum_{i=0}^{n-1} \Delta_{\tilde{f}} \circ f^i, \quad \forall n \in \mathbb{N}.$$

We have the following elementary but fundamental result about the **displacement function** of a lift of an orientation-preserving circle homeomorphism:

Proposition 5.3. It holds

$$\operatorname{osc} \Delta_{\tilde{f}} \leqslant 1, \quad \forall \tilde{f} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T}),$$

where the **oscillation of a function** $\varphi \in C^0(\mathbb{T}, \mathbb{R})$ is given by $\operatorname{osc} \varphi := \max_{x \in \mathbb{T}} \varphi(x) - \min_{y \in \mathbb{T}} \varphi(y)$.

Proof. Let us suppose that there is $\tilde{f} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ such that $\operatorname{osc} \Delta_{\tilde{f}} > 1$. Then, sine $\Delta_{\tilde{f}}$ is $\mathbb{Z}$-periodic, there are $x_m, x_M \in \mathbb{R}$ with $0 < x_M - x_m < 1$ and such that $\Delta_{\tilde{f}}(x_m) = \min_{\mathbb{T}} \Delta_{\tilde{f}}$ and $\Delta_{\tilde{f}}(x_M) = \max_{\mathbb{T}} \Delta_{\tilde{f}}$. Then, we have that

$$\tilde{f}(x_M) - \tilde{f}(x_m) = \Delta_{\tilde{f}}(x_M) - \Delta_{\tilde{f}}(x_m) + x_M - x_m > 1,$$

and hence, by continuity of $\tilde{f}$, there are two points $x_1, x_2 \in (x_m, x_M)$ such that $\tilde{f}(x_1) = \tilde{f}(x_2)+1$, and this implies that $f\big(\pi(x_1)\big) = \pi\big(\tilde{f}(x_1)\big) = \pi\big(\tilde{f}(x_2)+1\big) = \pi\big(\tilde{f}(x_2)\big) = f\big(\pi(x_2)\big)$. On the other hand, since $|x_2 - x_1| < x_M - x_m < 1$, we have $\pi(x_1) \neq \pi(x_2)$, which contradicts the fact that f is injective. $\square$

Proposition 5.3 lets us define the most important dynamical invariant of orientation-preserving circle homeomorphisms:

Theorem 5.4. Given any $\tilde{f} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$, there is a number $\rho(\tilde{f}) \in \mathbb{R}$ such that

$$\rho(\tilde{f}) = \lim_{n \to +\infty} \frac{\tilde{f}^n(x) - x}{n} = \lim_{n \to +\infty} \frac{\Delta_{\tilde{f}^n}(x)}{n}, \quad \forall x \in \mathbb{R}, \tag{5}$$

where the limit is uniform in $x \in \mathbb{R}$. On the other hand, for every $m, n \in \mathbb{Z}$, it holds

$$\rho(\tilde{f}^n \circ T_m) = \rho(\tilde{f}^n) + m = n\rho(\tilde{f}) + m,$$

where $T_m \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ denotes the translation given by $T_m(x) = x + m$, for every $x \in \mathbb{R}$.

In order to prove Theorem 5.4, we need the so called **subadditive lemma**:

Lemma 5.5. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that there is a real constant $C > 0$ satisfying

$$|a_{m+n} - a_m - a_n| \leqslant C, \quad \forall m, n \in \mathbb{N}.$$

Then, the following limit exists

$$\rho := \lim_{n \to +\infty} \frac{a_n}{n},$$

and it holds $|a_n - n\rho| \leqslant C$, for every $n \in \mathbb{N}$.

Exercise 5.5. Prove Lemma 5.5.

Proof of Theorem 5.4. Let us start proving that the limit (5) exists for $x = 0$. In fact, this is an easy consequence of Proposition 5.3 and Lemma 5.5. Indeed, by Proposition 5.3, we have that

$$\begin{aligned}\left|\Delta_{\tilde{f}^{m+n}}(0) - \Delta_{\tilde{f}^m}(0) - \Delta_{\tilde{f}^n}(0)\right| &= \left|\tilde{f}^{m+n}(0) - \tilde{f}^m(0) - \tilde{f}^n(0)\right| \\ &= \left|\Delta_{\tilde{f}^n}\left(\tilde{f}^m(0)\right) - \Delta_{\tilde{f}^n}(0)\right| \leqslant \operatorname{osc}\Delta_{\tilde{f}^n} \leqslant 1,\end{aligned}$$

for every $m, n \in \mathbb{N}$. So, by Lemma 5.5, there is $\rho(\tilde{f}) \in \mathbb{R}$ such that $\lim_{n\to+\infty} \frac{\Delta_{\tilde{f}^n}(0)}{n} = \rho(\tilde{f})$.

Invoking again Proposition 5.3, we have that

$$\left|\frac{\Delta_{\tilde{f}^n}(x)}{n} - \frac{\Delta_{\tilde{f}^n}(0)}{n}\right| \leqslant \frac{\operatorname{osc}\Delta_{\tilde{f}^n}}{n} \leqslant \frac{1}{n},$$

and hence, the limit is uniform in $x \in \mathbb{R}$.

Finally, given any $m, n \in \mathbb{Z}$, by Exercise 5.4, we have that

$$\begin{aligned}\rho(\tilde{f}^n \circ T_m) &= \lim_{k\to+\infty} \frac{(\tilde{f}^n \circ T_m)^k(x) - x}{k} = \lim_{k\to+\infty} \frac{\tilde{f}^{nk}(x) - x + mk}{k} \\ &= m + \lim_{k\to+\infty} n\frac{\tilde{f}^{nk}(x) - x}{nk} = n\rho(\tilde{f}) + m.\end{aligned}$$

□

We will need the following elementary but fundamental property of the rotation number:

Proposition 5.6. If $\tilde{f} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ and $\rho(\tilde{f})$ denotes the rotation number of $\tilde{f}$, then for each $n \in \mathbb{N}$ there exists $x_n \in \mathbb{R}$ such that

$$\tilde{f}^n(x_n) = x_n + n\rho(\tilde{f}).$$

Proof. Let us fix $n \in \mathbb{N}$. Suppose there is no $x \in \mathbb{R}$ such that $\tilde{f}^n(x) = x + n\rho(\tilde{f})$. Then, let us assume that $\tilde{f}^n(x) > x + n\rho(\tilde{f})$. Since $\Delta_{\tilde{f}^n} = \tilde{f}^n - id_{\mathbb{R}}$ is $\mathbb{Z}$-periodic, there exists $\varepsilon > 0$ such that $\tilde{f}^n(x) \geqslant x + n\rho(\tilde{f}) + \varepsilon$, for every $x \in \mathbb{R}$. So, by Proposition 5.3 we have that

$$\begin{aligned}\rho(\tilde{f}) &= \lim_{k\to+\infty} \frac{\tilde{f}^{nk}(x) - x}{nk} = \lim_{k\to+\infty} \frac{\sum_{i=0}^{k-1} \Delta_{\tilde{f}^n}\left(\tilde{f}^{ni}(x)\right)}{nk} \\ &\geqslant \lim_{k\to+\infty} \frac{nk\rho(\tilde{f}) + k\varepsilon}{nk} = \rho(\tilde{f}) + \frac{\varepsilon}{n},\end{aligned}$$

which is a contradiction. □

We have two important consequences Theorem 5.4 and Proposition 5.6:

Corollary 5.7 (Existence of periodic points)**.** If $f \in \mathrm{Homeo}_+(\mathbb{T})$ and $\mathrm{Per}(f) \neq \varnothing$, then $\rho(\tilde{f}) \in \mathbb{Q}$, for every lift $\tilde{f}$ of f. Conversely, if $\rho(\tilde{f}) = p/q \in \mathbb{Q}$, with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ coprime, then there is $\tilde{x} \in \mathbb{R}$ such that $\tilde{f}^q(\tilde{x}) = \tilde{x} + p$, and every periodic point for f has prime period q.

Proof. If $x \in \mathbb{T}$ is a periodic point for f with prime period $q \in \mathbb{N}$, then for every $\tilde{x} \in \pi^{-1}(x)$, there is $p \in \mathbb{Z}$ such that $\tilde{f}^q(\tilde{x}) = \tilde{x} + p$. By Proposition 5.3 we know that $\tilde{f}$ commutes with every integer translation, and hence, the number p does not depend on the choice of $\tilde{x} \in \pi^{-1}(x)$. So, if we compute the rotation number of $\tilde{f}$ using the point $\tilde{x}$, we get that $\rho(\tilde{f}) = \lim_{n\to+\infty} \frac{\tilde{f}^{nq}(\tilde{x})-\tilde{x}}{nq} = \lim_{n\to+\infty} \frac{np}{nq} = p/q$.

Conversely, if $\rho(\tilde{f}) = p/q \in \mathbb{Q}$, with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ coprime, then by Proposition 5.6 there is $\tilde{x} \in \mathbb{R}$ such that $\tilde{f}^q(\tilde{x}) = \tilde{x} + p$. So, the point $x := \pi(\tilde{x})$ is a periodic point for f, where q is a multiple of the prime period of x. By the previous argument, if the prime period of x were strictly smaller than q, the rotation number of $\tilde{f}$ would be a rational number with denominator strictly smaller than q, which contradicts the fact that p/q is written in irreducible form. □

As an immediate consequence of Proposition 5.3, Theorem 5.4 and Proposition 5.6, we have the following fundamental result

Corollary 5.8. Given any $\tilde{f} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$, it holds

$$\left|\tilde{f}^n(x) - x - n\rho(\tilde{f})\right| \leqslant 1, \quad \forall x \in \mathbb{R},\ \forall n \in \mathbb{Z}.$$

As a consequence of this estimates, we get that the rotation number function $\rho\colon \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T}) \to \mathbb{R}$ is continuous, when $\widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ is endowed with the uniform convergence topology.

Proof. First observe that for every $n \in \mathbb{N}$, there exists $x_n \in \mathbb{R}$ such that $\Delta_{\tilde{f}^n}(x_n) = n\rho(\tilde{f})$. On the other hand, by Proposition 5.3, we have that $\operatorname{osc} \Delta_{\tilde{f}^n} \leqslant 1$, for every n. So, we conclude $\left|\tilde{f}^n(x) - x - n\rho(\tilde{f})\right| = \left|\Delta_{\tilde{f}^n}(x) - \Delta_{\tilde{f}^n}(x_n)\right| \leqslant \operatorname{osc} \Delta_{\tilde{f}^n} \leqslant 1$, for every $n \in \mathbb{N}$. The case, $n < 0$ easily follows by applying the previous estimates to the homeomorphism $\tilde{f}^{-1}$.

In order to prove that the rotation number function is continuous, let us fix $\tilde{f} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ and $\varepsilon > 0$. Let n be a positive integer such that $3/n < \varepsilon$. Then, let us choose a real number $\delta > 0$ such that

$$\left|\tilde{f}^j(x) - \tilde{g}^j(x)\right| < 1, \quad \forall x \in \mathbb{R},\ \forall j \in \{1, 2, \ldots, n\},$$

for every $\tilde{g} \in \widetilde{\mathrm{Homeo}}_{0+}(\mathbb{T})$ such that $\left|\tilde{g}(x) - \tilde{f}(x)\right| < \delta$, for every $x \in \mathbb{R}$. For such a $\tilde{g}$, we have that

$$\begin{aligned}\left|\rho(\tilde{f}) - \rho(\tilde{g})\right| &\leqslant \left|\rho(\tilde{f}) - \frac{\tilde{f}^n(x) - x}{n}\right| + \left|\frac{\tilde{f}^n(x) - x}{n} - \frac{\tilde{g}^n(x) - x}{n}\right| + \left|\frac{\tilde{g}^n(x) - x}{n} - \rho(\tilde{g})\right| \\ &\leqslant \frac{1}{n} + \frac{1}{n}\left|\tilde{f}^n(x) - \tilde{g}^n(x)\right| + \frac{1}{n} < \frac{3}{n} < \varepsilon,\end{aligned}$$

and the continuity of the rotation number function is proved. □

References

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